

AVENZA

Final AI Dataset + Model Training Plan

Research/prototype scope: multimodal infant movement and cardiorespiratory event analysis.

Safety boundary: This is an educational/non-clinical prototype plan. It does not establish clinical safety or diagnostic performance. Public datasets have different populations, modalities and labels; their scores must not be merged into one claimed neonatal accuracy.

1. Final Avenza Architecture

CAMERA → infant/chest ROI → movement/respiratory-motion features → temporal video encoder
MAX30102 → raw RED/IR PPG → preprocessing → PPG/HR/SpO₂ features → temporal physiological encoder
(optional future ECG) → ECG temporal encoder
→ late-fusion MLP → event probability → signal-quality gate → state machine → dashboard

2. Final Dataset Set

Dataset	Contents	Avenza role
PICS – Preterm Infant Cardio-Respiratory Signals	200 preterm infants; synchronized ECG + respiration; about 20-70 days physiological foundation;	ECG/respiration features and bradycardia/apnoea/ECG apnoea, etc. models; to have one set
Apnea-ECG	70 long ECG records; 35 learning + 35 test; human-expert-developed ECG apnoea, etc. models; to have one set	ECG/respiration features and bradycardia/apnoea/ECG apnoea, etc. models; to have one set
babyPose	16 depth videos of preterm infants in NICU; 1000 frames (~100 s); pose only; 12 annotated limb joints.	pose only; 12 annotated limb joints.
Infant Movement Assessment	Infant video/pose resources, normative movement database and/or infant movement data presentation and temporal modelling.	pose only; 12 annotated limb joints.
Avenza synchronized dataset	Your camera + MAX30102 PPG, synchronized timestamps, final labelled waveforms; signal-quality/seamless validation.	pose only; 12 annotated limb joints.

3. Official Dataset Links

- [PICS – PhysioNet](#)
- [Apnea-ECG – PhysioNet](#)
- [babyPose – University of Macerata / Zenodo information](#)
- [Infant Movement Assessment – GitHub](#)

4. Model Choice

Video movement: pose/ROI or optical-flow time series → **1D CNN + GRU**.

ECG apnea research: ECG → preprocessing/RR features + waveform → **1D CNN + GRU**.

MAX30102: raw RED/IR PPG + HR/SpO₂ + quality features → **1D CNN + GRU**.

Fusion: modality embeddings → **late-fusion MLP**.

Safety layer: dedicated signal-quality/artifact classifier + rule/state logic.

Why: these are temporal physiological/visual signals. CNN captures local patterns; GRU captures temporal context. A large Transformer is not automatically more accurate, especially when the labelled dataset is small.

5. Video Movement Plan

babyPose is a neonatal movement/pose dataset, not an apnea-labelled dataset. It contains 16 depth videos recorded in NICU practice, each 1000 frames (~100 s), with 12 annotated limb joints. Use it to pretrain/validate movement representation, not to claim apnea accuracy.

Recommended pipeline:
Video → infant/chest ROI → pose or optical flow → temporal smoothing → 1D CNN → GRU → movement/respiratory-motion embedding.

For Avenza, chest/abdomen respiratory motion is more relevant to apnea than generic limb movement. Therefore, final respiratory-motion behaviour must be validated on Avenza's own camera setup.

6. Physiological / ECG Plan

PICS contains synchronized ECG and respiration from 10 preterm infants in a NICU. ECG is generally 500 Hz (250 Hz compound ECG for infants 1 and 5); respiration is generally 50 Hz. R-peak annotations were visually inspected; respiration peaks were algorithmically extracted and not manually vetted. It also provides bradycardia onset annotations.

Apnea-ECG contains 70 long ECG recordings, with 35 learning and 35 test records, plus human-expert-derived apnea annotations. Eight records also contain chest/abdominal respiratory effort, airflow and SpO₂. It is a sleep-apnea dataset and is not neonatal.

Therefore, use PICS for neonatal physiological context and Apnea-ECG for ECG-apnea methodology. Do not report either dataset's performance as Avenza's neonatal clinical performance.

7. MAX30102 — Actual Avenza Hardware

Your current hardware has a MAX30102, which provides optical RED/IR PPG rather than ECG. The final deployed branch should therefore be trained/evaluated on MAX30102 raw RED/IR recordings collected using the same sensor/optical setup used by the prototype.

Recommended inputs: raw RED/IR waveform; pulse/HR features; SpO₂ only when signal quality is adequate; PPG morphology; signal-quality indicators; timestamp/packet continuity.

If ECG is added later, it becomes an additional modality rather than replacing the MAX30102 branch.

8. Training Pipeline

- 1) Audit sampling rates, timestamps, missing segments and annotations.
- 2) Preprocess and detect artifacts.
- 3) Create synchronized temporal windows.
- 4) Split by complete subject/infant — never randomly split overlapping windows from the same subject across train/validation/test.
- 5) Establish a feature-based baseline such as XGBoost/Random Forest.
- 6) Train 1D CNN + GRU branches.
- 7) Train late-fusion MLP.
- 8) Calibrate probability and apply event persistence/recovery logic.

9. Fusion + Event Logic

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VIDEO ENCODER → video embedding
PPG/ECG ENCODER → physiological embedding
→ concatenate → MLP → calibrated event probability
→ signal-quality gate → event state machine
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Suggested research states: NORMAL → SUSPECTED → EVENT → RECOVERY, with separate SENSOR_ERROR and MOTION_ARTIFACT states. 'No movement + HR=0' must not automatically mean apnea; signal loss can create the same observation.

10. Evaluation — Optimize More Than Accuracy

Metric	What it tells us
Sensitivity / Recall	How many labelled events are detected.
Specificity	How well non-events are rejected.
Precision / PPV	How many predicted events are labelled events.
F1	Balance of precision and recall.
AUROC / AUPRC	Threshold-independent discrimination; AUPRC is useful for rare events.
False alarms/hour	Operational false-alert burden.
Event-level sensitivity	Whether complete events are detected, not just individual windows.
Detection latency	Delay from reference event onset to detection.
Artifact rejection	Ability to avoid calling sensor/motion failure an event.

11. Final Locked Recommendation

Video: babyPose + Infant Movement Assessment for movement/pose pretraining, then Avenza-specific synchronized recordings. Main temporal model: **1D CNN + GRU**.

Physiology: PICS for neonatal ECG/respiration context; Apnea-ECG for ECG-apnea methodology. Main model: **1D CNN + GRU**.

Actual Avenza sensor: MAX30102 raw RED/IR PPG + HR/SpO₂ + signal quality. Main model: **1D CNN + GRU**.

Final multimodal model: late-fusion **MLP** over the modality embeddings, followed by a signal-quality gate and event state machine.

Highest-priority new data: synchronized Avenza camera + MAX30102 recordings with precise timestamps and independently defined research labels. This determines transfer to the actual hardware.

12. Source Links

[PICS official PhysioNet page](#)

[Apnea-ECG official PhysioNet page](#)

[babyPose repository page](#)

[Infant Movement Assessment repository](#)